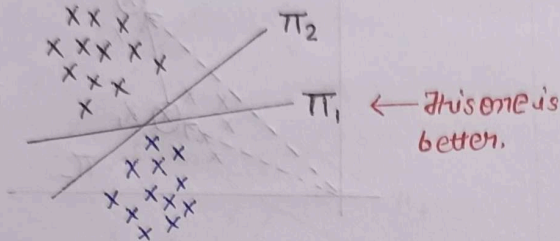


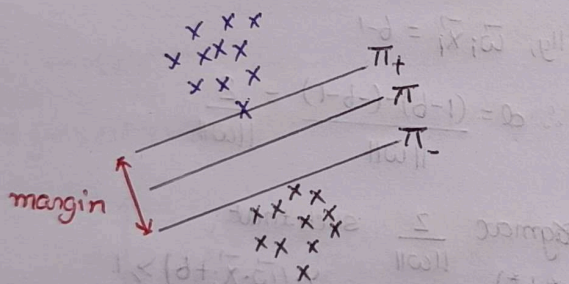
S.V.M.

SVM Geometric Intuition:

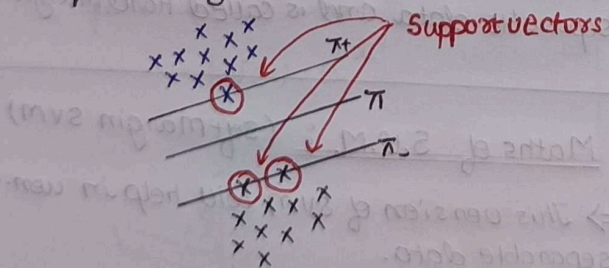
$\Rightarrow$ we will go for the hyperplane which be far from both classes, creating maximum gap or margin.



$\Rightarrow$ Basically, we go for margin maximising hyperplane. But how? we move both above and below and stop the moment we touches the data points. For above we called on π_+ and for below we call on π_- . Then we calc. distance b/w π_+ and π_- . We do it for every hyperplane and select the maximum one.

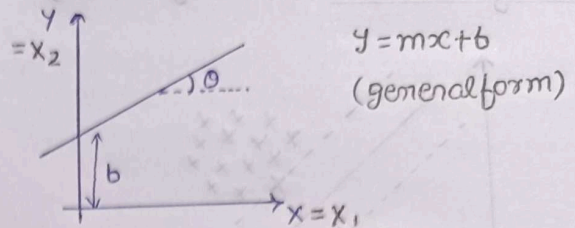


$\Rightarrow$ what are these support vectors? These are points responsible for creating π_+ and π_- .



Equation of a Hyperplane:

$\Rightarrow$ Equation of a line:



$y = mx + b$
(general form)

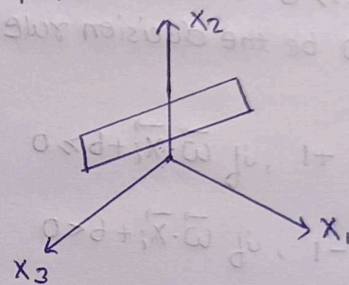
$$ax + by + c = 0 \Rightarrow y = -\frac{a}{b}x - \frac{c}{b}$$

$$\therefore ax_1 + bx_2 + c = 0$$

$$\Rightarrow \omega_1 x_1 + \omega_2 x_2 + \omega_0 = 0 \rightarrow \text{for 2D}$$

||y for 3D,

$$\omega_1 x_1 + \omega_2 x_2 + \omega_3 x_3 + \omega_0 = 0$$



||y for n-dimension:

$$\omega_1 x_1 + \omega_2 x_2 + \omega_3 x_3 + \dots + \omega_n x_n + \omega_0 = 0$$

Also,

$$\text{if } \omega = [\omega_1, \omega_2, \dots, \omega_n]$$

$$X = [x_1, x_2, \dots, x_n]$$

$$\therefore \omega^T X + \omega_0 = 0 \rightarrow \text{eqn of line in any dimension}$$

$$\therefore y = mx + b$$

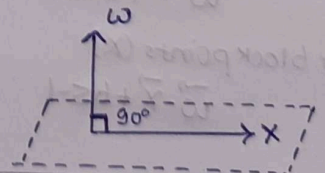
$$\therefore b = -\frac{\omega_0}{\omega_2}$$

$$\therefore \text{if plane passes from origin, then } \omega_0 = 0$$

$$\therefore \omega^T X + \omega_0 = 0 \Rightarrow \omega^T X = 0$$

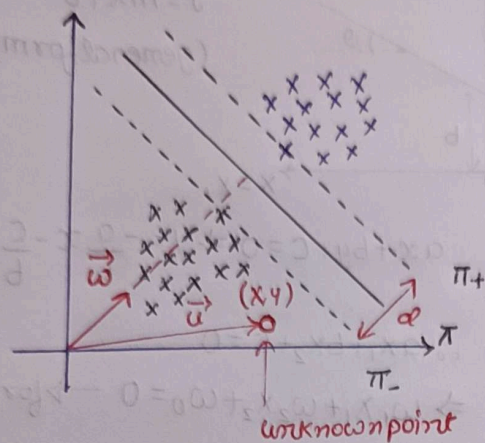
$$\therefore \omega^T X = \omega \cdot X = \|\omega\| \|X\| \cos \theta = 0$$

$$\Rightarrow \theta = 90^\circ$$



Maths of S.V.M. (Hard margin SVM)

⇒



$$\because \vec{\omega} \cdot \vec{u} \geq c$$

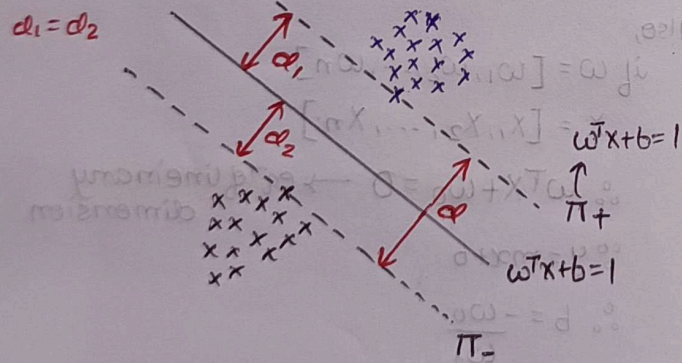
$$\Rightarrow \vec{\omega} \cdot \vec{u} - c \geq 0 \Rightarrow \vec{\omega} \cdot \vec{u} + b \geq 0$$

This $\vec{\omega} \cdot \vec{u} + b \geq 0$ be the decision rule.

$\because$ for any $\vec{x}_i$

$$\hat{y} = \begin{cases} +1, & \text{if } \vec{\omega} \cdot \vec{x}_i + b \geq 0 \\ -1, & \text{if } \vec{\omega} \cdot \vec{x}_i + b < 0 \end{cases}$$

⇒ Let eqⁿ of π_+ be $\vec{\omega}^T \vec{x} + b = 1$ and π_- be $\vec{\omega}^T \vec{x} + b = -1$.



Now, for all blue points (x), the equation be:

$$\vec{\omega} \cdot \vec{x} + b \geq 1$$

lly, for black points (x)

$$\vec{\omega} \cdot \vec{x} + b \leq -1$$

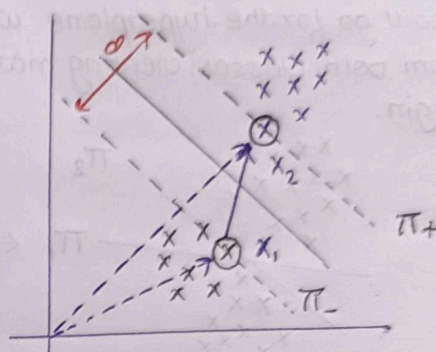
Let, $Y_i = 1$ be label for blue points

$$\because Y_i (\vec{\omega} \cdot \vec{x}_i + b) \geq 1$$

for all support vectors it be

$$Y_i (\vec{\omega} \cdot \vec{x}_i + b) = 1$$

Now,



$$\because d = (\vec{x}_2 - \vec{x}_1) \cdot \frac{\vec{\omega}}{\|\vec{\omega}\|}$$

$$= \frac{\vec{x}_2 \cdot \vec{\omega} - \vec{x}_1 \cdot \vec{\omega}}{\|\vec{\omega}\|}$$

$$\because Y_i (\vec{\omega} \cdot \vec{x}_i + b) = 1$$

$$\Rightarrow 1 (\vec{\omega} \cdot \vec{x}_2 + b) = 1 \Rightarrow \vec{\omega} \cdot \vec{x}_2 = 1 - b$$

$$\text{lly, } \vec{\omega} \cdot \vec{x}_1 = b - 1$$

$$\because d = \frac{(1-b) - (b-1)}{\|\vec{\omega}\|} = \frac{2}{\|\vec{\omega}\|}$$

$$\because \text{argmax}_{(\vec{\omega}^*, b^*)} \frac{2}{\|\vec{\omega}\|} \text{ such that } Y_i (\vec{\omega} \cdot \vec{x}_i + b) \geq 1$$

Since, this whole formulation is for only linear separable data and is called Hard margin SVM.

Maths of S.V.M. (Soft margin SVM)

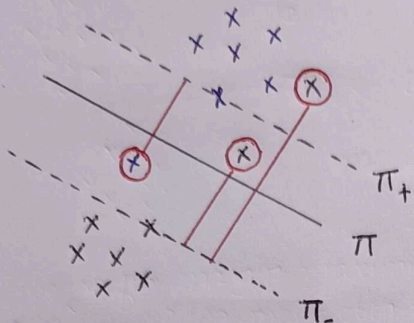
⇒ This version of SVM will help in non-linear separable data.

$$\Rightarrow \because \max f(x) \Leftrightarrow \min \frac{1}{f(x)}$$

$\therefore \arg \min_{(\omega^*, b^*)} \frac{\|\omega\|^2}{2} + c \sum_{i=1}^n \xi_i$
hyperparameters

where, $\xi_i = 0$ for correctly classified points

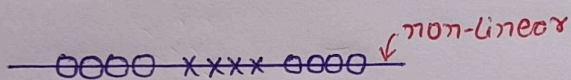
$\Rightarrow \arg \min_{(\omega^*, b^*)} \text{Margin error} + \text{classification error}$



Kernel Trick SVM

$\Rightarrow$ It's used to convert any non-linear data into linear data by increasing its dimensions.

e.g.,



$\Rightarrow$ The function which do so is called kernel.

e.g.,

